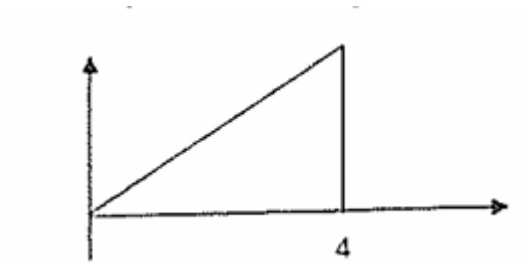
Question bank

1	Apply overlap-add and overlap save method to filter long data sequence $x(n) = \{3, 0, -2, 0, 2, 1, 0, -2, -1, 2,3\}$ with the filter given by $h(n) = \{2, 1, 3\}$. Verify your result using matrix method.
2	What is windowing technique in filter design? Design a HPF for the following desired frequency response given by $$H_d(e^{jw}) = \begin{cases} 0 & \frac{-\pi}{3} \le w \le \frac{\pi}{3} \\ e^{-j4w} & \frac{\pi}{3} < \lvert w \rvert \le \pi \end{cases}$$ Determine the filter coefficients $h(n)$ if the window function is defined using rectangular window.
3	What is twiddle factor? Write its two properties.
4	For a sequence x(n)= {1 3 5 7} Find its DFT X(K). And from the X(K) obtain its IDFT x(n).
5	For the given H(z) $$H(z) = \frac{3 + 5z^{-1} + \frac{1}{3}z^{-2}}{2 + \frac{7}{3}z^{-1} - \frac{5}{3}z^{-2}}$$ Find the difference equation which represent the above system. Represent the system by using DF-I and DF-II representation. Compare between DF1 and DF2 structure in terms of computational complexity.
6	Determine the Fourier Transform of the following signal $\mathbf{x(t) = rect(\frac{t}{6})}$.
7	For two given sequences x1(n)={1,3,2,5} and x2(n)={3,2,4,2}, Find circular convolution between x1(n) an x2(n). Verify circular convolution property of DFT. Also find Linear convolution using circular convolution.
8	Discuss S-plane and z-Plane mapping.
9	Discuss about the relation between DTFT and Z-transform.
10	The system function of a causal LTI system is $H(z) = \frac{1-z^{-1}}{1+\frac{3}{4}z^{-1}}$. The input to this system is $x(n) = \left(\frac{1}{3}\right)^n u(n) + u(-n-1)$. Find (a) Impulse response of the system $h(n)$ (b) Find the output $y(n)$ (c) Is the system stable
11	Suppose that we want to design a discrete time LTI system with the property that if the input $x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u(n-1)$ then the output $y(n) =$

	$\left(\frac{1}{3}\right)^n u(n)$. (a) Find the impulse response and frequency response of a discrete time LTI system that has the foregoing property (b) Find the difference equation relating $x(n)$ and $y(n)$ that characterizes the system.
12	Design a LPF for the following desired frequency response given by $$H_d(e^{jw}) = \begin{cases} e^{-j4w} & \frac{-\pi}{3} \le w \le \frac{\pi}{3} \\ 0 & \frac{\pi}{3} < \lvert w \rvert \le \pi \end{cases}$$ Determine the filter coefficients if the window function is defined using rectangular window.
13	Find the exponential Fourier series representation of a sawtooth wave having a period of 4 units and maximum amplitude of 3 units (whose one period is given below) 
14	Check whether the system $y(n) = \cos[x(n)]$ is (a) static/dynamic (b) Linear/Nonlinear © Causal/ Non-causal d. Time variant/ Time invariant e. Stable/ unstable
15	Find the Fourier transform of the following signals (a) $x(t) = sgn(t)$ (b) X(t)=u(t)
16	Test the stability of the LTI systems whose impulse response is $h(n) = 0.3^n u(n) + 0.4^n u(n)$
17	For an LTI system the impulse response is given as $h(n) = 3\delta(n+1) - 2\delta(n-1) + 2\delta(n-2)$. Estimate the response of the system for the input

	$x(n) = 3\delta(n+1) + 2\delta(n) + 2\delta(n-1)$ using concentric circle method.
18	Consider a Causal LTI system with frequency response $H(w) = \frac{1}{jw+4}$. For a particular input $x(t)$ the system is observed to produce the output $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$. Determine the input to the system.